

RONSHEE CHAWLA

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EDUCATION

- **University of Texas at Austin (UT)**
PhD in Electrical and Computer Engineering August 2017 – Present
MS in Electrical and Computer Engineering August 2017 – May 2019
Advisor: Prof. *Sanjay Shakkottai*
Current GPA: 4.0/4.0
- **California Institute of Technology (Caltech)** September 2016 – August 2017
Graduate Student in Electrical Engineering
GPA: 3.7/4.0
- **Indian Institute of Technology (IIT) Indore, India** July 2011 – May 2015
B. Tech. in Electrical Engineering
Advisor: Prof. *Prabhat K. Upadhyay*
GPA: 9.89/10.0

RESEARCH INTERESTS

- Design and analysis of algorithms for decentralized/distributed machine learning
- Reinforcement learning theory and algorithms
- Bandit algorithms

SELECT RESEARCH PROJECTS

- *The Gossiping Insert–Eliminate Algorithm for Multi–Agent Bandits*
 - Considered a multi-agent multi-armed bandit setup, wherein agents exchange recommendations through pairwise gossip to find the best action under a communication budget
 - Proposed two novel decentralized algorithms, which greatly reduce regret by a factor of order number of agents, despite limited communication
- *Multi–Agent Low–Dimensional Linear Bandits*
 - Examined a new multi-agent linear bandit setting with low-dimensional subspace side information
 - Proposed a novel decentralized algorithm utilizing collaboration between agents, resulting in much lower finite-time regret per agent

TECHNICAL SKILLS

- **Programming languages:** C++, Python
- **Softwares:** MATLAB

PUBLICATIONS (* denotes equal contribution)

- *Multi–Agent Low–Dimensional Linear Bandits*
R. Chawla, Abishek Sankararaman, Sanjay Shakkottai
IEEE Transactions on Automatic Control, 2022
- *The Gossiping Insert–Eliminate Algorithm for Multi–Agent Bandits*
R. Chawla*, Abishek Sankararaman*, Ayalvadi Ganesh and Sanjay Shakkottai
International Conference on Artificial Intelligence and Statistics (AISTATS) 2020

- *Outage Performance and Location Optimization for Traffic-Aware Two-Way Relaying with Direct Link*
Suneel Yadav, R. Chawla and Prabhat K. Upadhyay
International Conference on Signal Processing and Communications (SPCOM) 2016
- *Outage Analysis of Cellular Two-Way Relaying with Spectrum Sharing in Nakagami- m Fading*
R. Chawla, M. Mounika Reddy and Prabhat K. Upadhyay
Proceedings of 18th International Symposium on Wireless Personal Multimedia Communications (WPMC) 2015

TEACHING EXPERIENCE

- Teaching Assistant for the class EE 351K *Probability, Statistics and Random Processes* at UT, Fall 2017.
- Teaching Assistant for the class EE 381V *Online Learning* at UT, Fall 2019.

TALKS

- *The Gossiping Insert-Eliminate Algorithm for Multi-Agent Bandits*
Invited talk at Vrbo, Austin, TX February 2020

WORK EXPERIENCE

- **Scientist-B** August 2015 – May 2016
Aeronautical Development Establishment (ADE), DRDO, Bengaluru, India
Flight Test Telecommand and Tracking Division
- **Internship** May 2014 – July 2014
RWTH Aachen University, Germany
Worked on automatic extraction of layer-specific data-flow configurations for automatic scheduling and mapping of kernels for the Layers architecture

HONORS AND ACHIEVEMENTS

- First rank in Electrical Engineering PhD Qualifying Exam at Caltech, January 2017.
- **President of India Gold Medal**, first rank in IIT Indore, class of 2015.
- DAAD WISE scholarship for summer internship at RWTH Aachen University, Germany, May 2014 – July 2014.
- Offered summer fellowship by Indian Academy of Sciences, May 2014 – July 2014 (declined).
- Academic Excellence Awards, IIT Indore, 2012 – 13, 2013 – 14 and 2014 – 15.

RELEVANT COURSEWORK

- **Graduate Coursework** (at UT): Probability and Stochastic Processes I, Convex Optimization, Advanced Probability, Special Topics in Machine Learning (Unsupervised Learning), Large Scale Optimization, Topics in Learning Theory, Fair Transparent Machine Learning, Analysis and Design of Communication Networks
- **Graduate Coursework** (at Caltech): Information Theory, Learning Systems (Machine Learning), Applied Linear Algebra, Random Matrices, Error-Correcting Codes, Mathematics of Signal Processing
- **Undergraduate Coursework**: Computer Programming, Signals and Systems, Digital Signal Processing, Advanced Signal Processing